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Hello World in Android Studio using GUI Components, Fonts and Colors

Expt. No: 01
Date:06/09/2021

Aim:

To implement hello world in android studio using GUI components, fonts and colours.

Software and Hardware Requirements:

Software:

1. Android Studio
2. Java

Hardware:

1. Preferably 8GB+ RAM Laptop/Desktop

Algorithm:

1. Open Android Studio and create a new project.
2. Select Empty Activity.
3. In the text attribute mention the text that has to be displayed.
4. Background specifies the background colour

Code:

MainActivity.java:

```
package com.example.guiapplication;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.graphics.Color;

public class MainActivity extends AppCompatActivity {

    int ch=1;
    float font=30;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        final TextView t= (TextView) findViewById(R.id.textView);
        Button b1= (Button) findViewById(R.id.button1);
```

```
        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                t.setTextSize(font);
                font = font + 5;
                if (font == 50)
                    font = 30;
            }
        });
        Button b2= (Button) findViewById(R.id.button2);
        b2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                switch (ch) {
                    case 1:
                        t.setTextColor(Color.RED);
                        break;
                    case 2:
                        t.setTextColor(Color.GREEN);
                        break;
                    case 3:
                        t.setTextColor(Color.BLUE);
                        break;
                    case 4:
                        t.setTextColor(Color.CYAN);
                        break;
                    case 5:
                        t.setTextColor(Color.YELLOW);
                        break;
                    case 6:
                        t.setTextColor(Color.MAGENTA);
                        break;
                }
                ch++;
                if (ch == 7)
                    ch = 1;
            }
        });
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/teal_200"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="30dp"
        android:gravity="center"
        android:text="Welcome to our Application!"
        android:textColor="@color/black"
        android:textSize="34sp"
        android:textStyle="bold"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintHorizontal_bias="0.417"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.156" />

    <Button
        android:id="@+id/button1"
        android:layout_width="301dp"
        android:layout_height="51dp"
        android:layout_margin="20dp"
        android:gravity="center"
        android:text="Change font size"
        android:textSize="25sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintHorizontal_bias="0.545"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.608" />

    <Button
        android:id="@+id/button2"
        android:layout_width="308dp"
        android:layout_height="52dp"
        android:layout_margin="20dp"
        android:gravity="center"
        android:text="Change color"
        android:textSize="25sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintHorizontal_bias="0.631"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.883" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Sample I/O:

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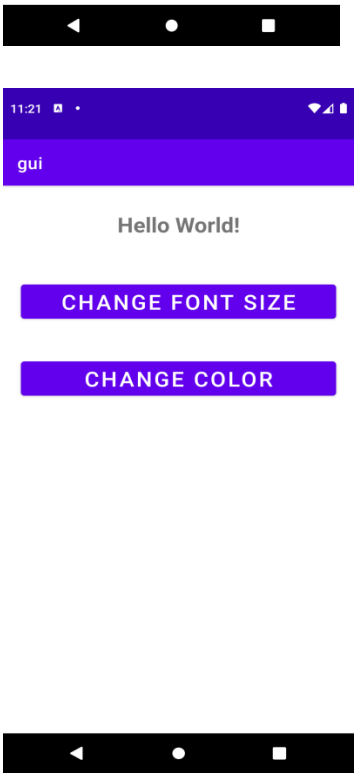

Result:

Thus, hello world was successfully implemented in Android Studio using GUI Components, fonts and colours.

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Layout Managers and Event Listeners

Expt. No: 02
Date:13/09/2021

Aim:

To develop a Simple Android Application that uses Layout Managers and Event Listeners.

Software and Hardware Requirements:

Software:

3. Android Studio
4. Java

Hardware:

1. Preferably 8GB+ RAM Laptop/Desktop

Algorithm:

1) Open Android Studio and create a new project.
2) Select Empty Activity.
3) In this application, the activity implements Constraint Layout, Linear Layout, and Relative Layout
4) There are various event listeners like onclick, ondoubleclick, onlongPress, onTouch events. In this activity onclick is implemented.

Code:

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="100dp">
        <TextView
            android:id="@+id/textView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_margin="30dp"
            android:text="Details Form"
            android:textSize="25sp"
```

```
        android:gravity="center"/>
</LinearLayout>

<GridLayout
    android:id="@+id/gridLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_marginTop="100dp"
    android:layout_marginBottom="200dp"
    android:columnCount="2"
    android:rowCount="3">
    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_row="0"
        android:layout_column="0"
        android:text="Name"
        android:textSize="20sp"
        android:gravity="center"/>

    <EditText
        android:id="@+id/editText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_row="0"
        android:layout_column="1"
        android:ems="10"/>

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_row="1"
        android:layout_column="0"
        android:text="Reg.No"
        android:textSize="20sp"
        android:gravity="center"/>

    <EditText
        android:id="@+id/editText2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_row="1"
        android:layout_column="1"
        android:inputType="number"
        android:ems="10"/>

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
```

```
            android:layout_height="wrap_content"
            android:layout_margin="10dp"
            android:layout_row="2"
            android:layout_column="0"
            android:text="Dept"
            android:textSize="20sp"
            android:gravity="center"/>

        <Spinner
            android:id="@+id/spinner"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_margin="10dp"
            android:layout_row="2"
            android:layout_column="1"
            android:spinnerMode="dropdown"/>

    </GridLayout>

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentBottom="true"
        android:layout_centerInParent="true"
        android:layout_marginBottom="150dp"
        android:text="Submit"/>

</RelativeLayout>
```

Activity_second.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context="com.example.devang.exno2.SecondActivity"
    android:orientation="vertical"
    android:gravity="center">

    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="New Text"
        android:textSize="30sp"/>

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="New Text"
```

```
        android:textSize="30sp"/>

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="New Text"
        android:textSize="30sp"/>

</LinearLayout>
```

SecondActivity.java:

```
package com.example.layoutmanager;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.widget.TextView;
import android.os.Bundle;

public class SecondActivity extends AppCompatActivity {
    TextView t1,t2,t3;

    String name,reg,dept;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        t1= (TextView) findViewById(R.id.textView1);
        t2= (TextView) findViewById(R.id.textView2);
        t3= (TextView) findViewById(R.id.textView3);

        //Getting the Intent
        Intent i = getIntent();

        //Getting the Values from First Activity using the Intent
received
        name=i.getStringExtra("name_key");
        reg=i.getStringExtra("reg_key");
        dept=i.getStringExtra("dept_key");

        //Setting the Values to Intent
        t1.setText(name);
        t2.setText(reg);
        t3.setText(dept);
    }
```

MainActivity.java:

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```
package com.example.layoutmanager;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Spinner;

public class MainActivity extends AppCompatActivity {

    //Defining the Views
    EditText e1,e2;
    Button bt;
    Spinner s;

    //Data for populating in Spinner
    String [] dept_array={"CSE","ECE","IT","Mech","Civil"};

    String name,reg,dept;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        //Referring the Views
        e1= (EditText) findViewById(R.id.editText);
        e2= (EditText) findViewById(R.id.editText2);

        bt= (Button) findViewById(R.id.button);

        s= (Spinner) findViewById(R.id.spinner);

        //Creating Adapter for Spinner for adapting the data from
array to Spinner
        ArrayAdapter adapter= new
ArrayAdapter(MainActivity.this,android.R.layout.simple_spinner_item,
dept_array);
        s.setAdapter(adapter);

        //Creating Listener for Button
        bt.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                //Getting the Values from Views(Edittext & Spinner)
                name=e1.getText().toString();
                reg=e2.getText().toString();
                dept=s.getSelectedItem().toString();

                //Intent For Navigating to Second Activity
                Intent i = new
```

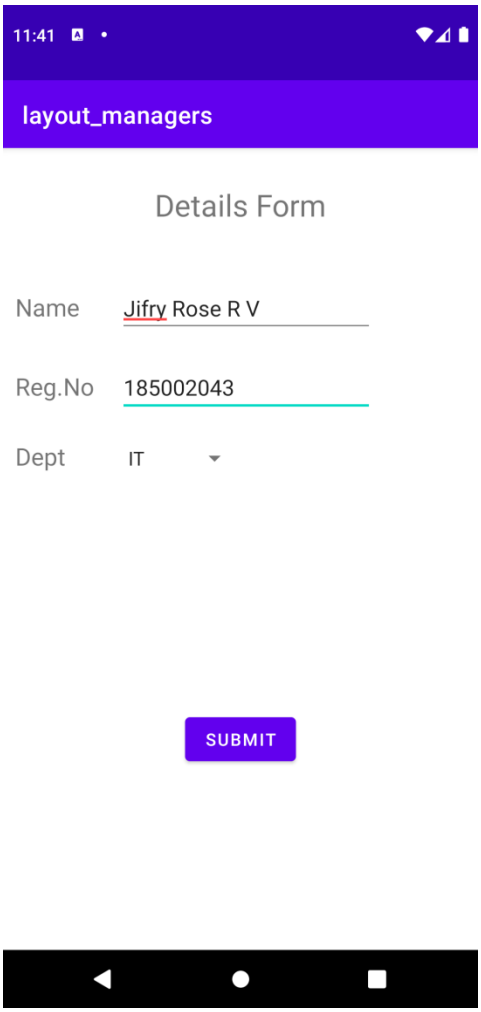
```
Intent(MainActivity.this,SecondActivity.class);

                //For Passing the Values to Second Activity
                i.putExtra("name_key", name);
                i.putExtra("reg_key",reg);
                i.putExtra("dept_key", dept);

                startActivity(i);

            }
        });
    }
}
```

Sample I/O:



Result:

Thus a Simple Android Application that uses Layout Managers and Event Listeners is developed and executed successfully.

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Simple Calculator Application

Exp No : 3
Date : 18/09/2021

Aim:

To implement a simple calculator in android studio.

Software and Hardware Requirements:

Software:

5. Android Studio
6. Java

Hardware:

2. Preferably 8GB+ RAM Laptop/Desktop

Algorithm:

1) Open Android Studio and then click on File -> New -> New project.
2) Then type the Application name as “calculator″ and click Next.
3) Then select the Empty Activity and click Next.
4) Finally click Finish.
5) The buttons required to implement the calculator are provided in the xml file.
6) The implementation of functions for the respective button clicks are specified in the java file.
7) When equal button is pressed the function implementation is done

Code:

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#8BC34A"
    android:backgroundTint="@android:color/darker_gray"
    tools:context=".MainActivity">

    <!-- Text View to display our basic heading of "calculator"-->
    <TextView
        android:layout_width="194dp"
        android:layout_height="43dp"
        android:layout_marginStart="114dp"
```

```
        android:layout_marginLeft="114dp"
        android:layout_marginTop="58dp"
        android:layout_marginEnd="103dp"
        android:layout_marginRight="103dp"
        android:layout_marginBottom="502dp"
        android:scrollbarSize="30dp"
        android:text=" Calculator"

android:textAppearance="@style/TextAppearance.AppCompat.Body1"
        android:textSize="30dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <!-- Edit Text View to input the values -->
    <EditText
        android:id="@+id/num1"
        android:layout_width="364dp"
        android:layout_height="28dp"
        android:layout_marginStart="72dp"
        android:layout_marginTop="70dp"
        android:layout_marginEnd="71dp"
        android:layout_marginBottom="416dp"
        android:background="@android:color/white"
        android:ems="10"
        android:hint="Number1(0)"
        android:inputType="number"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <!-- Edit Text View to input 2nd value-->
    <EditText
        android:id="@+id/num2"
        android:layout_width="363dp"
        android:layout_height="30dp"
        android:layout_marginStart="72dp"
        android:layout_marginTop="112dp"
        android:layout_marginEnd="71dp"
        android:layout_marginBottom="374dp"
        android:background="@android:color/white"
        android:ems="10"
        android:hint="number2(0)"
        android:inputType="number"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <!-- Text View to display result -->
    <TextView
        android:id="@+id/result"
        android:layout_width="356dp"
```

```
    android:layout_height="71dp"
    android:layout_marginStart="41dp"
    android:layout_marginTop="151dp"
    android:layout_marginEnd="48dp"
    android:layout_marginBottom="287dp"
    android:background="@android:color/white"
    android:text="result"
    android:textColorLink="#673AB7"
    android:textSize="25sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<!-- A button to perform 'sum' operation -->
<Button
    android:id="@+id/sum"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginTop="292dp"
    android:layout_marginEnd="307dp"
    android:layout_marginBottom="263dp"
    android:backgroundTint="@android:color/holo_red_light"
    android:onClick="doSum"
    android:text="+"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<!-- A button to perform subtraction operation. -->
<Button
    android:id="@+id/sub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="210dp"
    android:layout_marginTop="292dp"
    android:layout_marginEnd="113dp"
    android:layout_marginBottom="263dp"
    android:backgroundTint="@android:color/holo_red_light"
    android:onClick="doSub"
    android:text="-"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<!-- A button to perform division. -->
<Button
    android:id="@+id/div"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="307dp"
    android:layout_marginTop="292dp"
```

```
    android:layout_marginEnd="16dp"
    android:layout_marginBottom="263dp"
    android:backgroundTint="@android:color/holo_red_light"
    android:onClick="doDiv"
    android:text="/"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<!-- A button to perform multiplication. -->
<Button
    android:id="@+id/mul"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginTop="356dp"
    android:layout_marginEnd="307dp"
    android:layout_marginBottom="199dp"
    android:backgroundTint="@android:color/holo_red_light"
    android:onClick="doMul"
    android:text="x"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<!-- A button to perform a modulus function. -->
<Button
    android:id="@+id/button"
    android:layout_width="92dp"
    android:layout_height="48dp"
    android:layout_marginStart="113dp"
    android:layout_marginTop="356dp"
    android:layout_marginEnd="206dp"
    android:layout_marginBottom="199dp"
    android:backgroundTint="@android:color/holo_red_light"
    android:onClick="doMod"
    android:text="%(mod)"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<!-- A button to perform a power function. -->
<Button
    android:id="@+id/pow"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="113dp"
    android:layout_marginTop="292dp"
    android:layout_marginEnd="210dp"
    android:layout_marginBottom="263dp"
    android:backgroundTint="@android:color/holo_red_light"
```

```
        android:onClick="doPow"
        android:text="n1^n2"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

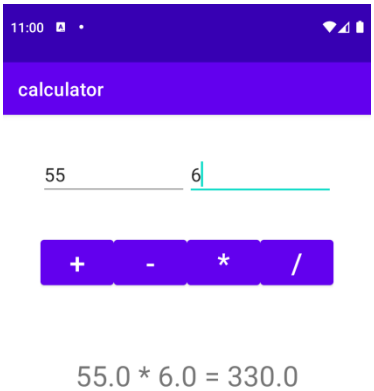
MainActivity_java:

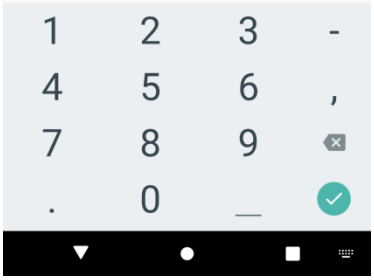
```
package com.example.calculator_sdk;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    EditText e1, e2;
    TextView t1;
    int num1, num2;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    // a public method to get the input numbers
    public boolean getNumbers() {

        // defining the edit text 1 to e1
        e1 = (EditText) findViewById(R.id.num1);

        // defining the edit text 2 to e2
        e2 = (EditText) findViewById(R.id.num2);

        // defining the text view to t1
        t1 = (TextView) findViewById(R.id.result);

        // taking input from text box 1
        String s1 = e1.getText().toString();

        // taking input from text box 2
        String s2 = e2.getText().toString();

        // condition to check if box is not empty
        if ((s1.equals(null) && s2.equals(null))
                || (s1.equals("") && s2.equals(""))) {

            String result = "Please enter a value";
```

```
        t1.setText(result);

        return false;
    } else {
        // converting string to int.
        num1 = Integer.parseInt(s1);

        // converting string to int.
        num2 = Integer.parseInt(s2);
    }

    return true;
}

// a public method to perform addition
public void doSum(View v) {

    // get the input numbers
    if (getNumbers()) {
        int sum = num1 + num2;
        t1.setText(Integer.toString(sum));
    }
}

// a public method to perform power function
public void doPow(View v) {

    // get the input numbers
    if (getNumbers()) {
        double sum = Math.pow(num1, num2);
        t1.setText(Double.toString(sum));
    }
}

// a public method to perform subtraction
public void doSub(View v) {

    // get the input numbers
    if (getNumbers()) {
        int sum = num1 - num2;
        t1.setText(Integer.toString(sum));
    }
}

// a public method to perform multiplication
public void doMul(View v) {

    // get the input numbers
    if (getNumbers()) {
        int sum = num1 * num2;
        t1.setText(Integer.toString(sum));
    }
}

// a public method to perform Division
```

```
    public void doDiv(View v) {

        // get the input numbers
        if (getNumbers()) {

            // displaying the text in text view assigned as t1
            double sum = num1 / (num2 * 1.0);
            t1.setText(Double.toString(sum));
        }
    }

    // a public method to perform modulus function
    public void doMod(View v) {

        // get the input numbers
        if (getNumbers()) {
            double sum = num1 % num2;
            t1.setText(Double.toString(sum));
        }
    }
}
```

Sample I/O:





Result:

Thus a Simple Android Application for Native Calculator is developed and executed successfully.

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Basic Graphical Primitives

Expt. No: 04
Date: 20/09/2021

Aim:

To implement basic graphical primitives in android studio.

Software and Hardware Requirements:

Software:

1. Android Studio
2. Java

Hardware:

1. Preferably 8GB+ RAM Laptop/Desktop

Algorithm:

1. Open Android Studio and create a new project.
2. Select Empty Activity.
3. Android provides a huge set of 2D-drawing APIs that allow you to create graphics.
4. The Bitmap (android.graphics.Bitmap) class represents a bitmap image.
5. A Canvas is an object that you can draw on by calling drawing commands. In this program, we have drawn a rectangle, circle, square and line using Canvas class.

Code:

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:id="@+id/imageView" />
</RelativeLayout>
```

MainActivity.java:

```
package com.example.graphicalapplication;

import android.os.Bundle;
import android.app.Activity;
import android.graphics.Bitmap;
import android.graphics.Canvas;
import android.graphics.Color;
```

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```
import android.graphics.Paint;
import android.graphics.drawable.BitmapDrawable;
import android.widget.ImageView;

public class MainActivity extends Activity
{
    @Override
    public void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        //Creating a Bitmap
        Bitmap bg = Bitmap.createBitmap(720, 1280,
Bitmap.Config.ARGB_8888);

        //Setting the Bitmap as background for the ImageView
        ImageView i = (ImageView) findViewById(R.id.imageView);
        i.setBackgroundDrawable(new BitmapDrawable(bg));

        //Creating the Canvas Object
        Canvas canvas = new Canvas(bg);

        //Creating the Paint Object and set its color & TextSize
        Paint paint = new Paint();
        paint.setColor(Color.RED);
        paint.setTextSize(50);

        //To draw a Rectangle
        canvas.drawText("Rectangle", 420, 150, paint);
        canvas.drawRect(400, 200, 650, 700, paint);

        //To draw a Circle
        canvas.drawText("Circle", 120, 150, paint);
        canvas.drawCircle(200, 350, 150, paint);

        //To draw a Square
        canvas.drawText("Square", 120, 800, paint);
        canvas.drawRect(50, 850, 350, 1150, paint);

        //To draw a Line
        canvas.drawText("Line", 480, 800, paint);
        canvas.drawLine(520, 850, 520, 1150, paint);
    }
}
```

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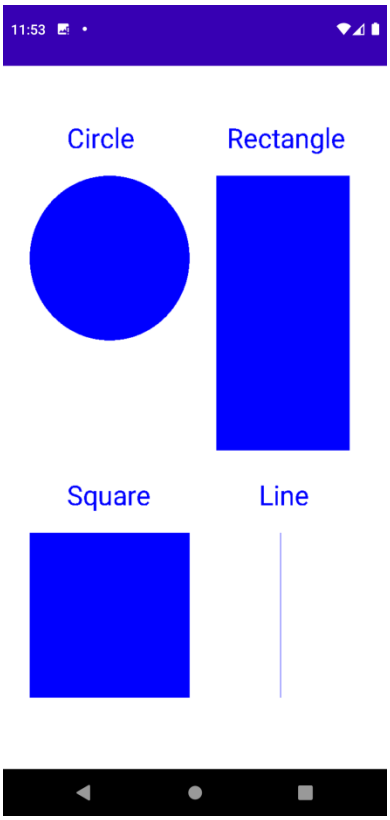
Sample I/O:



Result:

Thus a Simple Android Application that draws basic Graphical Primitives on the screen is developed and executed successfully.

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Database

Expt. No: 05
Date: 25/09/2021

Aim:
To develop a simple Android Application using a database.

Software and Hardware Requirements:
Software:
1. Android Studio
2. Java

Hardware:
1. Preferably 8GB+ RAM Laptop/Desktop

Algorithm:
1. Open Android Studio and create a new project.
2. Select Empty Activity.
3. SQLiteDatabase has methods to create, delete, execute SQL commands, and perform other common database management tasks.
4. The Builder class has several methods. This program makes use of the following:
   - setCancelable(boolean flag): Sets whether this dialog is cancelable with the BACK key.
   - setTitle(char Sequence): Makes the string passed as the title.
5. Event Listeners are used to track mouse clicks

Code:

MainActivity.java
```
package com.example.database;

import android.os.Bundle;

import android.app.Activity;
import android.app.AlertDialog.Builder;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;

import android.os.Bundle;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
```

```
import android.widget.EditText;

public class MainActivity extends Activity implements OnClickListener
{
   EditText Rollno,Name,Marks;
   Button Insert,Delete,Update,View,ViewAll;
   SQLiteDatabase db;
   /** Called when the activity is first created. */
   @Override
   public void onCreate(Bundle savedInstanceState)
   {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       Rollno=(EditText)findViewById(R.id.Rollno);
       Name=(EditText)findViewById(R.id.Name);
       Marks=(EditText)findViewById(R.id.Marks);
       Insert=(Button)findViewById(R.id.Insert);
       Delete=(Button)findViewById(R.id.Delete);
       Update=(Button)findViewById(R.id.Update);
       View=(Button)findViewById(R.id.View);
       ViewAll=(Button)findViewById(R.id.ViewAll);

       Insert.setOnClickListener(this);
       Delete.setOnClickListener(this);
       Update.setOnClickListener(this);
       View.setOnClickListener(this);
       ViewAll.setOnClickListener(this);

       // Creating database and table
       db=openOrCreateDatabase("StudentDB", Context.MODE_PRIVATE, null);
       db.execSQL("CREATE TABLE IF NOT EXISTS student(rollno VARCHAR,name
VARCHAR,marks VARCHAR);");
   }
   public void onClick(View view)
   {
       // Inserting a record to the Student table
       if(view==Insert)
       {
           // Checking for empty fields
           if(Rollno.getText().toString().trim().length()==0||
                   Name.getText().toString().trim().length()==0||
                   Marks.getText().toString().trim().length()==0)
           {
               showMessage("Error", "Please enter all values");
               return;
           }
           db.execSQL("INSERT INTO student
VALUES('"+Rollno.getText()+"','"+Name.getText()+
                   "','"+Marks.getText()+"');");
           showMessage("Success", "Record added");
           clearText();
       }
```

```
        // Deleting a record from the Student table
        if(view==Delete)
        {
            // Checking for empty roll number
            if(Rollno.getText().toString().trim().length()==0)
            {
                showMessage("Error", "Please enter Rollno");
                return;
            }
            Cursor c=db.rawQuery("SELECT * FROM student WHERE
rollno='"+Rollno.getText()+"'", null);
            if(c.moveToFirst())
            {
                db.execSQL("DELETE FROM student WHERE
rollno='"+Rollno.getText()+"'");
                showMessage("Success", "Record Deleted");
            }
            else
            {
                showMessage("Error", "Invalid Rollno");
            }
            clearText();
        }
        // Updating a record in the Student table
        if(view==Update)
        {
            // Checking for empty roll number
            if(Rollno.getText().toString().trim().length()==0)
            {
                showMessage("Error", "Please enter Rollno");
                return;
            }
            Cursor c=db.rawQuery("SELECT * FROM student WHERE
rollno='"+Rollno.getText()+"'", null);
            if(c.moveToFirst()) {
                db.execSQL("UPDATE student SET name='" + Name.getText() +
"',marks='" + Marks.getText() +
                        "' WHERE rollno='"+Rollno.getText()+"'");
                showMessage("Success", "Record Modified");
            }
            else {
                showMessage("Error", "Invalid Rollno");
            }
            clearText();
        }
        // Display a record from the Student table
        if(view==View)
        {
            // Checking for empty roll number
            if(Rollno.getText().toString().trim().length()==0)
            {
                showMessage("Error", "Please enter Rollno");
                return;
```

```
            }
            Cursor c=db.rawQuery("SELECT * FROM student WHERE
rollno='"+Rollno.getText()+"'", null);
            if(c.moveToFirst())
            {
                Name.setText(c.getString(1));
                Marks.setText(c.getString(2));
            }
            else
            {
                showMessage("Error", "Invalid Rollno");
                clearText();
            }
        }
        // Displaying all the records
        if(view==ViewAll)
        {
            Cursor c=db.rawQuery("SELECT * FROM student", null);
            if(c.getCount()==0)
            {
                showMessage("Error", "No records found");
                return;
            }
            StringBuffer buffer=new StringBuffer();
            while(c.moveToNext())
            {
                buffer.append("Rollno: "+c.getString(0)+"\n");
                buffer.append("Name: "+c.getString(1)+"\n");
                buffer.append("Marks: "+c.getString(2)+"\n\n");
            }
            showMessage("Student Details", buffer.toString());
        }
    }
    public void showMessage(String title,String message)
    {
        Builder builder=new Builder(this);
        builder.setCancelable(true);
        builder.setTitle(title);
        builder.setMessage(message);
        builder.show();
    }
    public void clearText()
    {
        Rollno.setText("");
        Name.setText("");
        Marks.setText("");
        Rollno.requestFocus();
    }
}
```

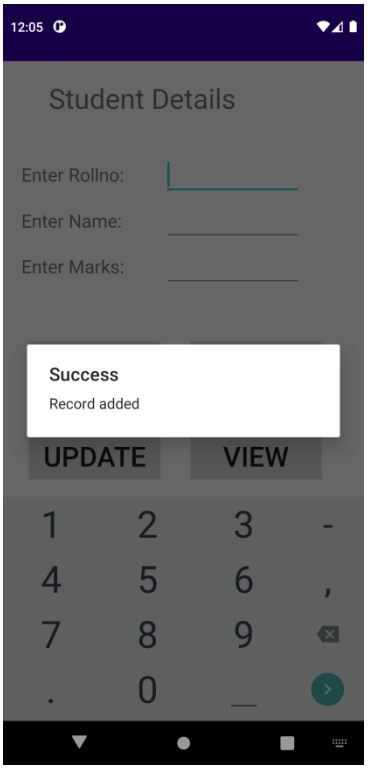
activity_main.xml:

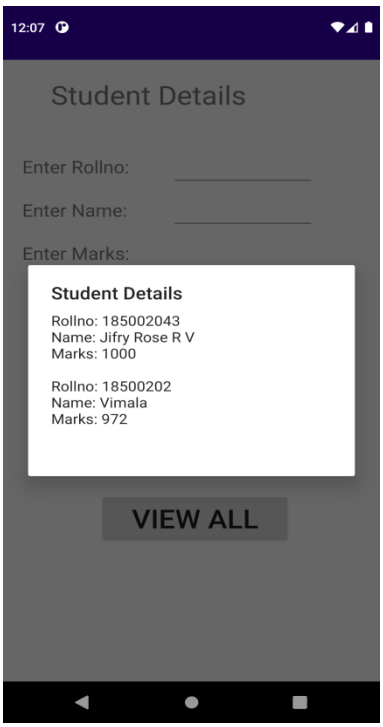
```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">
   <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_x="50dp"
       android:layout_y="20dp"
       android:text="Student Details"
       android:textSize="30sp" />

   <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_x="20dp"
       android:layout_y="110dp"
       android:text="Enter Rollno:"
       android:textSize="20sp" />

   <EditText
       android:id="@+id/Rollno"
       android:layout_width="150dp"
       android:layout_height="wrap_content"
       android:layout_x="175dp"
       android:layout_y="100dp"
       android:inputType="number"
       android:textSize="20sp" />

   <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_x="20dp"
       android:layout_y="160dp"
       android:text="Enter Name:"
       android:textSize="20sp" />

   <EditText
       android:id="@+id/Name"
       android:layout_width="150dp"
       android:layout_height="wrap_content"
       android:layout_x="175dp"
       android:layout_y="150dp"
       android:inputType="text"
       android:textSize="20sp" />

   <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_x="20dp"
       android:layout_y="210dp"
       android:text="Enter Marks:"
```

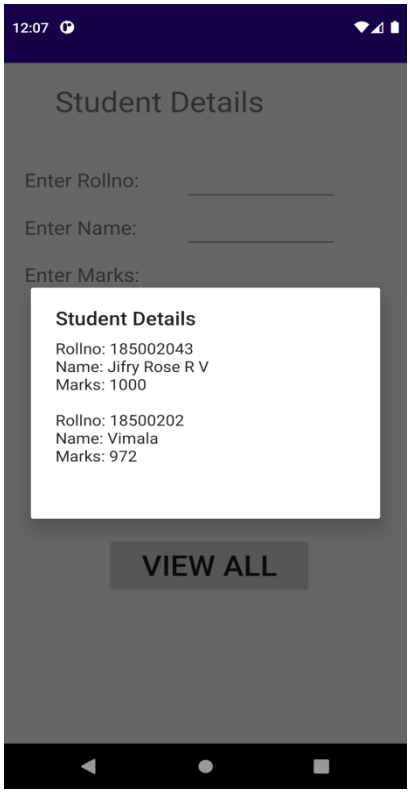
```
    android:textSize="20sp" />

<EditText
    android:id="@+id/Marks"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="175dp"
    android:layout_y="200dp"
    android:inputType="number"
    android:textSize="20sp" />

<Button
    android:id="@+id/Insert"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="300dp"
    android:text="Insert"
    android:textSize="30dp" />

<Button
    android:id="@+id/Delete"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="300dp"
    android:text="Delete"
    android:textSize="30dp" />

<Button
    android:id="@+id/Update"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="400dp"
    android:text="Update"
    android:textSize="30dp" />

<Button
    android:id="@+id/View"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="400dp"
    android:text="View"
    android:textSize="30dp" />

<Button
    android:id="@+id/ViewAll"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:layout_x="100dp"
    android:layout_y="500dp"
```

```
        android:text="View All"
        android:textSize="30dp" />

</AbsoluteLayout>
```

Sample I/O:







Result:

Thus, a simple application using a database has been successfully implemented using android studio.

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Multithreading

Expt.No:06
Date:27/092021

Aim:

To develop an Android Application that implements Multithreading.

Software and Hardware Requirements:

Software:

1. Android Studio
2. Java

Hardware:

1. Preferably 8GB+ RAM Laptop/Desktop

Algorithm:

1. A thread is a thread of execution in a program. The Java Virtual Machine allows an application to have multiple threads of execution running concurrently.
2. When a Java Virtual Machine starts up, there is usually a single non-daemon thread (which typically calls the method named main of some designated class). The Java Virtual Machine continues to execute threads until either of the following occurs:
   i. · The exit method of class Runtime has been called and the security manager has permitted the exit operation to take place.
   ii. · All threads that are not daemon threads have died, either by returning from the call to the run method or by throwing an exception that propagates beyond the run method.
3. Using the runnable instance of the thread module, we multi-thread the process of loading two images of the India map.
4. On selecting the first button, one type of image will be displayed and on selecting the other button, the second image will be displayed immediately

Code:

MainActivity.java:

```
package com.example.multithreading;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
public class MainActivity extends AppCompatActivity
```

```
{
      ImageView img;
      Button bt1,bt2;
      @Override
      protected void onCreate(Bundle savedInstanceState)
      {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      bt1 = (Button)findViewById(R.id.button);
      bt2= (Button) findViewById(R.id.button2);
      img = (ImageView)findViewById(R.id.imageView);

      bt1.setOnClickListener(new View.OnClickListener()
      {
            @Override
            public void onClick(View v)
            {
                  new Thread(new Runnable()
                  {
                  @Override
                  public void run()
                     {
                        img.post(new Runnable()
                        {
                              @Override
                              public void run()
                              {
                              img.setImageResource(R.drawable.pic1);
                              }
                        });
                  }
                  }).start();
            }
      });

      bt2.setOnClickListener(new View.OnClickListener()
      {
            @Override
             public void onClick(View v)
            {
                  new Thread(new Runnable()
                  {
                  @Override
                  public void run()
                  {
                        img.post(new Runnable()
                      {
                              @Override
                              public void run()
                              {
                              img.setImageResource(R.drawable.pic2);
                              }
                        });
                  }
                  }).start();
            }
      });
      }
}
```

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activitymain.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical" >

      <ImageView
      android:id="@+id/imageView"
      android:layout_width="250dp"
      android:layout_height="250dp"
      android:layout_margin="50dp"
      android:layout_gravity="center" />

      <Button
      android:id="@+id/button"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_margin="5dp"
      android:layout_gravity="center"
      android:text="Load Image 1" />

      <Button
      android:id="@+id/button2"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_margin="5dp"
      android:layout_gravity="center"
      android:text="Load image 2" />

</LinearLayout>
```

Sample I/O:

Result:

Thus, an Android Application that implements Multithreading has been successfully implemented.

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SD Card

Expt. No:07
Date:04/10/2021

Aim:

To develop an Android Application that writes data to the SD Card.

Algorithm:

1. Create a new flutter project
2. Create a text field with a controller.
3. Create a button, that when pressed, saved the text written on the text field to the sd card.
4. Get directory path from path provider package
5. Write the contents to the preferred file.

Code:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="SD Card App"
        android:textAppearance="@style/TextAppearance.AppCompat.Display1"
        android:textColor="#007565"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintHorizontal_bias="0.497"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.064" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="36dp"
        android:layout_marginEnd="108dp"
        android:text="Write contents into a file"
        android:textAppearance="@style/TextAppearance.AppCompat.Medium"
        android:textColor="#00BFA5"
```

```
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/title"
    android:layout_marginRight="108dp" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="80dp"
    android:layout_marginEnd="150dp"
    android:text="File contents:"
    android:textAppearance="@style/TextAppearance.AppCompat.Medium"
    android:textColor="#304FFE"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/title"
    android:layout_marginRight="152dp" />

<TextView
    android:id="@+id/textView3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="180dp"
    android:layout_marginEnd="150dp"
    android:text="File Name:"
    android:textAppearance="@style/TextAppearance.AppCompat.Medium"
    android:textColor="#304FFE"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/title"
    android:layout_marginRight="256dp" />

<EditText
    android:id="@+id/content"
    android:layout_width="166dp"
    android:layout_height="44dp"
    android:layout_marginTop="10dp"
    android:layout_marginEnd="40dp"
    android:ems="10"
    android:gravity="start|top"
    android:inputType="textMultiLine"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/textView2"
    android:layout_marginRight="40dp" />

<EditText
    android:id="@+id/filename"
    android:layout_width="163dp"
    android:layout_height="44dp"
    android:layout_marginTop="80dp"
    android:layout_marginEnd="60dp"
    android:ems="10"
    android:gravity="start|top"
    android:inputType="textMultiLine"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/textView2"
    android:layout_marginRight="60dp" />



<Button

    android:id="@+id/writebtn"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:layout_marginEnd="156dp"
        android:backgroundTint="#00BFA5"
        android:text="Write File"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/filename"
        android:layout_marginRight="156dp" />



    <Button
        android:id="@+id/btnLoad"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="100dp"
        android:layout_marginEnd="136dp"
        android:backgroundTint="#00BFA5"
        android:text="Load From File"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/filename"
        android:layout_marginRight="136dp" />



</androidx.constraintlayout.widget.ConstraintLayout>
```

activity_read.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ReadFile">



    <EditText
        android:id="@+id/filename2"
        android:layout_width="163dp"
        android:layout_height="40dp"
        android:layout_marginTop="28dp"
        android:layout_marginEnd="52dp"
        android:ems="10"
        android:gravity="start|top"
        android:inputType="textMultiLine"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView6"
        android:layout_marginRight="52dp" />


    <TextView
        android:id="@+id/textView4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
    android:layout_marginTop="40dp"
    android:layout_marginEnd="152dp"
    android:text="File contents:"
    android:textAppearance="@style/TextAppearance.AppCompat.Medium"
    android:textColor="#304FFE"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/loadbtn"
    android:layout_marginRight="152dp" />

<TextView
    android:id="@+id/title2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="44dp"
    android:layout_marginEnd="108dp"
    android:text="SD Card App"
    android:textAppearance="@style/TextAppearance.AppCompat.Display1"
    android:textColor="#007565"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    android:layout_marginRight="108dp" />

<TextView
    android:id="@+id/textView5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="100dp"
    android:layout_marginTop="28dp"
    android:text="File Name:"
    android:textAppearance="@style/TextAppearance.AppCompat.Medium"
    android:textColor="#304FFE"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/textView6"
    android:layout_marginLeft="100dp" />

<Button
    android:id="@+id/loadbtn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="36dp"
    android:layout_marginEnd="148dp"
    android:backgroundTint="#00BFA5"
    android:text="LOAD FILE"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/filename2"
    android:layout_marginRight="148dp" />

<TextView
    android:id="@+id/textView6"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:layout_marginEnd="104dp"
    android:text="Read contents From File"
    android:textAppearance="@style/TextAppearance.AppCompat.Medium"
    android:textColor="#00BFA5"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/title2"
    android:layout_marginRight="104dp" />

<TextView
```

```
        android:id="@+id/tvLoad"
        android:layout_width="283dp"
        android:layout_height="116dp"
        android:layout_marginTop="32dp"
        android:layout_marginEnd="52dp"
        android:textAppearance="@style/TextAppearance.AppCompat.Body1"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/textView4"
        android:layout_marginRight="52dp" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java:

```
import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;

import android.content.DialogInterface;

import android.content.Intent;

import android.content.pm.PackageManager;

import android.os.Build;

import android.os.Bundle;

import android.os.Environment;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.TextView;

import android.widget.Toast;


import java.io.BufferedReader;

import java.io.File;

import java.io.FileNotFoundException;

import java.io.FileOutputStream;

import java.io.FileReader;

import java.io.IOException;

import java.io.OutputStreamWriter;
```

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```
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import android.content.DialogInterface;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.os.Environment;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import java.io.BufferedReader;
import java.io.File;
import java.io.FileNotFoundException;
import java.io.FileOutputStream;
import java.io.FileReader;
import java.io.IOException;
import java.io.OutputStreamWriter;

public class MainActivity extends AppCompatActivity {
    // Declare the View object references
    Button btnSave, btnLoad;
    EditText content, filenamev;
    TextView tvLoad;
    // Define some String variables, initialized with empty string
```

```
String filepath = "";
String filename = "";
String fileContent = "";

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    btnSave = findViewById(R.id.writebtn);
    btnLoad = findViewById(R.id.btnLoad);
    content = findViewById(R.id.content);
    filenamev = findViewById(R.id.filename);
    filepath = "NewDirectory";

    if(!isExternalStorageAvailableForRW()){
        btnSave.setEnabled(false);
    }

    btnSave.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {

            fileContent = content.getText().toString().trim();
            filename = filenamev.getText().toString().trim();

            // Check for Storage Permission
            if(isStoragePermissionGranted()){
```

```
                    if(!fileContent.equals("")){

                        File myExternalFile = new
File(getExternalFilesDir(filepath), filename);

                        FileOutputStream fos = null;

                        try {

                            fos = new
FileOutputStream(myExternalFile);

                            fos.write(fileContent.getBytes());

                            fos.close();

                        } catch (FileNotFoundException e) {

                            e.printStackTrace();

                        } catch (IOException e) {

                            e.printStackTrace();

                        }



                        content.setText("");

                        filenamev.setText("");

                        // Show a Toast message to inform the user
that the operation has been successfully completed.

                        Toast.makeText(MainActivity.this, "File
saved to SD card.", Toast.LENGTH_SHORT).show();

                    } else{

                        // If the Text field is empty show
corresponding Toast message

                        Toast.makeText(MainActivity.this, "Text
field can not be empty.", Toast.LENGTH_SHORT).show();

                    }

                }



            }
```

```
        });


        btnLoad.setOnClickListener(new View.OnClickListener() {

            @Override

            public void onClick(View view) {

                Intent myIntent = new Intent(MainActivity.this,
ReadFile.class);

                MainActivity.this.startActivity(myIntent);

            }

        });

    }


    public boolean isStoragePermissionGranted() {

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.M) {

            if
(checkSelfPermission(android.Manifest.permission.WRITE_EXTERNAL_STOR
AGE)

                    == PackageManager.PERMISSION_GRANTED) {

                //Permission is granted

                return true;

            } else {

                //Permission is revoked

                ActivityCompat.requestPermissions(this, new
String[]{android.Manifest.permission.WRITE_EXTERNAL_STORAGE}, 1);

                return false;

            }

        }

        else {

            //permission is automatically granted on sdk<23 upon
installation

            //Permission is granted
```

```
            return true;

        }

    }



    private boolean isExternalStorageAvailableForRW() {

        // Check if the external storage is available for read and
write by calling

        // Environment.getExternalStorageState() method. If the
returned state is MEDIA_MOUNTED,

        // then you can read and write files. So, return true in
that case, otherwise, false.

        String extStorageState =
Environment.getExternalStorageState();

        if(extStorageState.equals(Environment.MEDIA_MOUNTED)){

            return true;

        }

        return false;

    }

}
```

ReadFile.java:

```
package com.example.sd;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.TextView;

import java.io.BufferedReader;

import java.io.File;
```

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```
import java.io.FileNotFoundException;

import java.io.FileReader;

import java.io.IOException;

public class ReadFile extends AppCompatActivity {

      Button  btnLoad;

      TextView tvLoad;

      EditText filenamev;

      String filename = "";

          String filepath = "NewDirectory";

      @Override

      protected void onCreate(Bundle savedInstanceState) {

                  super.onCreate(savedInstanceState);

                  setContentView(R.layout.activity_read_file);

                  btnLoad = findViewById(R.id.loadbtn);

                  filenamev = findViewById(R.id.filename2);

                  tvLoad = findViewById(R.id.tvLoad);
btnLoad.setOnClickListener(new View.OnClickListener() {

@Override
public void onClick(View view) {

filename = filenamev.getText().toString().trim();

FileReader fr = null;

File myExternalFile = new File(getExternalFilesDir(filepath),
filename);

StringBuilder stringBuilder = new StringBuilder();

try {

fr = new FileReader(myExternalFile);
BufferedReader br = new BufferedReader(fr);

String line = br.readLine();

while(line != null){
```

```
stringBuilder.append(line).append('\n');

   line = br.readLine();

  }

} catch (FileNotFoundException e) {

  e.printStackTrace();

} catch (IOException e) {

     e.printStackTrace();
} finally {

 String fileContents = stringBuilder.toString();

                                        tvLoad.setText(fileContents);

                                  }

                        }

                 });

      }

}
```

AndroidManifest.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<manifest xmlns:android="http://schemas.android.com/apk/res/android"

      package="com.example.sd">



      <uses-permission
android:name="android.permission.WRITE_EXTERNAL_STORAGE" />

      <uses-permission
android:name="android.permission.READ_EXTERNAL_STORAGE" />



      <application

      android:allowBackup="true"

      android:icon="@mipmap/ic_launcher"

      android:label="@string/app_name"
```

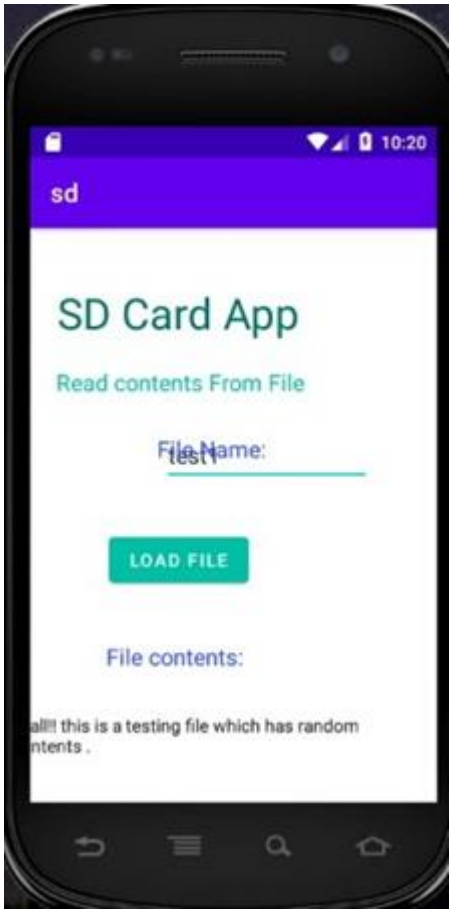
```
      android:roundIcon="@mipmap/ic_launcher_round"

      android:supportsRtl="true"

      android:theme="@style/Theme.Sd">

      <activity

            android:name=".ReadFile"

            android:exported="true" />

      <activity

            android:name=".MainActivity"

            android:exported="true">

            <intent-filter>

                  <action android:name="android.intent.action.MAIN" />



                  <category
android:name="android.intent.category.LAUNCHER" />

            </intent-filter>

      </activity>

      </application>



</manifest>
```

themes.xml:

```
<resources xmlns:tools="http://schemas.android.com/tools">

      <!-- Base application theme. -->

      <style name="Theme.Sd"
parent="Theme.MaterialComponents.DayNight.DarkActionBar">

      <!-- Primary brand color. -->

      <item name="colorPrimary">@color/purple_500</item>

      <item name="colorPrimaryVariant">@color/purple_700</item>

      <item name="colorOnPrimary">@color/white</item>

      <!-- Secondary brand color. -->
```

```
    <item name="colorSecondary">@color/teal_200</item>

    <item name="colorSecondaryVariant">@color/teal_700</item>

    <item name="colorOnSecondary">@color/black</item>

    <!-- Status bar color. -->

    <item name="android:statusBarColor"
tools:targetApi="l">?attr/colorPrimaryVariant</item>

    <!-- Customize your theme here. -->

    </style>

</resources>
```

Sample I/O:



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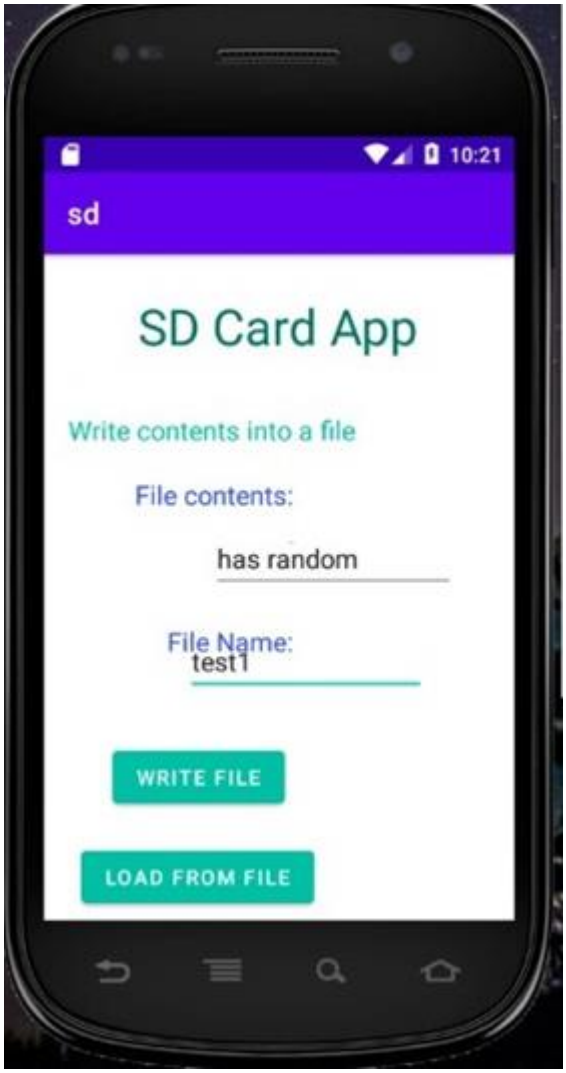

Result:

Thus, an Android Application that writes data to the SD Card was successfully implemented.

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Alert

Expt.No:08
Date:11/10/2021

Aim:

To develop an Android Application that creates an alert upon receiving a message.

Software and Hardware Requirements:

Software:

1. Android Studio
2. Java

Hardware:

1. Preferably 8GB+ RAM Laptop/Desktop

Algorithm:

1) Open Android Studio and then click on File -> New -> New project.
2) Then type the Application name as “alert″ and click Next.
3) Then select the Empty Activity and click Next.
4) In this application, a java class is created that makes use of BroadCastReceiver and SmsManager to get the sender number of the SMS along with the text that was sent.
5) These details are given as a toast message alert when the SMS is received by the device. This application uses the SMS permission to SEND, RECEIVE and READ SMS.

Code:

MainActivity.java:

```
package com.example.alert;

import android.app.Notification;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity{
    Button notify;
    EditText e;
    @Override
```

```
    public void onCreate(Bundle savedInstanceState){
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        notify= (Button) findViewById(R.id.button);
        e= (EditText) findViewById(R.id.editText);
        notify.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v){
                Intent intent = new Intent(MainActivity.this,
SecondActivity.class);
                PendingIntent pending =
PendingIntent.getActivity(MainActivity.this, 0, intent, 0);
                Notification noti = new
Notification.Builder(MainActivity.this).setContentTitle("New
Message").setContentText(e.getText().toString()).setSmallIcon(R.mipmap.ic_l
auncher).setContentIntent(pending).build();
                NotificationManager manager = (NotificationManager)
getSystemService(NOTIFICATION_SERVICE);
                noti.flags |= Notification.FLAG_AUTO_CANCEL;
                manager.notify(0, noti);
            }
        });
    }
}
```

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="10dp"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Message"
        android:textSize="30sp" />

    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:singleLine="true"
        android:textSize="30sp" />
    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="30dp"
        android:layout_gravity="center"
        android:text="Notify"
        android:textSize="30sp"/>
</LinearLayout>
```

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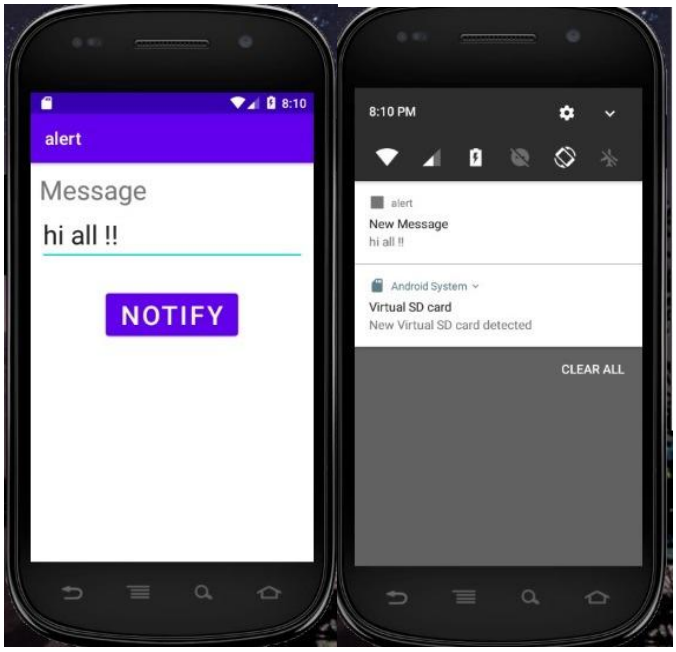
OUTPUT:



RESULT:

Thus Android Application that creates an alert upon receiving a message is developed and executed successfully.

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Alarm

Expt.No:09
Date:18/10/2021

Aim:
To develop an Android Application that creates an alert upon receiving a message.

Software and Hardware Requirements:
Software:
1. Android Studio
2. Java

Hardware:
1. Preferably 8GB+ RAM Laptop/Desktop

Algorithm:
1) Open Android Studio and then click on File -> New -> New project.
2) Then type the Application name as “alert″ and click Next.
3) Then select the Empty Activity and click Next.
4) In the xml file, we have added two items ‘TimePicker’ and ‘ToggleButton’. TimePicker is used to capture the alarm time and ToggleButton is added to set the alarm on or off.
5) In MainActivity.java class the onToggleClicked( ) method is implemented in which the current hour and the minute is set using the calendar.
6) Create a new java class named “AlarmReceiver.java” at the same place where MainActivity.java class resides. In this class the onReceive() method is implemented. Here we have added vibration functionality and a default ringtone that starts to vibrate and ring when the alarm time is scheduled.
7) Go to the “values” folder first then choose the colors.xml file. In the colors.xml file, you can keep colors of your choice as many as you want to use in your app.
8) Go to the “values” folder first then choose the themes.xml file. In the theme.xml file, we have used “Theme.AppCompat.Light.DarkActionBar” which is a light theme with a dark ActionBar
9) Go to the “AndroidManifest.xml” file. A BroadcastReceiver is registered in AndroidManifest.xml by adding a receiver section after the application section is over.

Code:

MainActivity.java:

```
package com.example.alarmclock;
import android.app.AlarmManager;
import android.app.PendingIntent;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.TimePicker;
import android.widget.Toast;
import android.widget.ToggleButton;

import androidx.appcompat.app.AppCompatActivity;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
    TimePicker alarmTimePicker;
    PendingIntent pendingIntent;
    AlarmManager alarmManager;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        alarmTimePicker = (TimePicker)
findViewById(R.id.timePicker);
        alarmManager = (AlarmManager)
getSystemService(ALARM_SERVICE);

    }

    // OnToggleClicked() method is implemented the time
functionality
    public void OnToggleClicked(View view) {
        long time;
        if (((ToggleButton) view).isChecked()) {
            Toast.makeText(MainActivity.this, "ALARM ON",
Toast.LENGTH_SHORT).show();
            Calendar calendar = Calendar.getInstance();

            // calender is called to get current time in hour and
minute
            calendar.set(Calendar.HOUR_OF_DAY,
alarmTimePicker.getCurrentHour());
            calendar.set(Calendar.MINUTE,
alarmTimePicker.getCurrentMinute());

            // using intent i have class AlarmReceiver class which
inherits
            // BroadcastReceiver
            Intent intent = new Intent(this, AlarmReceiver.class);

            // we call broadcast using pendingIntent
```

```
            pendingIntent = PendingIntent.getBroadcast(this, 0,
intent, 0);

            time = (calendar.getTimeInMillis() -
(calendar.getTimeInMillis() % 60000));
            if (System.currentTimeMillis() > time) {
                // setting time as AM and PM
                if (calendar.AM_PM == 0)
                    time = time + (1000 * 60 * 60 * 12);
                else
                    time = time + (1000 * 60 * 60 * 24);
            }
            // Alarm rings continuously until toggle button is
turned off
            alarmManager.setRepeating(AlarmManager.RTC_WAKEUP, time,
10000, pendingIntent);
            // alarmManager.set(AlarmManager.RTC_WAKEUP,
System.currentTimeMillis() + (time * 1000), pendingIntent);
        } else {
            alarmManager.cancel(pendingIntent);
            Toast.makeText(MainActivity.this, "ALARM OFF",
Toast.LENGTH_SHORT).show();
        }
    }
}
```

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <!--Added Time picker just to pick the alarm time-->
    <!--gravity is aligned to center-->
    <TimePicker
        android:id="@+id/timePicker"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center" />

    <!--Added Toggle Button to set the alarm on or off-->
    <!--ByDefault toggleButton is set to false-->
    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:layout_margin="20dp"
```

```
        android:checked="false"
        android:onClick="OnToggleClicked" />

    <!--"OnToggleClicked" method will be implemented in
MainActivity.java -->

</LinearLayout>
```

AlarmReceiver.java

```
package com.example.alarmclock;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.media.Ringtone;
import android.media.RingtoneManager;
import android.net.Uri;
import android.os.Build;
import android.os.Vibrator;
import android.widget.Toast;

import androidx.annotation.RequiresApi;

public class AlarmReceiver extends BroadcastReceiver {
    @RequiresApi(api = Build.VERSION_CODES.Q)
    @Override
    // implement onReceive() method
    public void onReceive(Context context, Intent intent) {

        // we will use vibrator first
        Vibrator vibrator = (Vibrator)
context.getSystemService(context.VIBRATOR_SERVICE);
        vibrator.vibrate(4000);

        Toast.makeText(context, "Alarm! Wake up! Wake up!",
Toast.LENGTH_LONG).show();
        Uri alarmUri =
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_ALARM);
        if (alarmUri == null) {
            alarmUri =
RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);
        }

        // setting default ringtone
        Ringtone ringtone = RingtoneManager.getRingtone(context,
alarmUri);

        // play ringtone
        ringtone.play();
```

```
    }
}
```

Themes.xml:

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Theme.Alarmclock"
parent="Theme.MaterialComponents.DayNight.DarkActionBar">
        <!-- Primary brand color. -->
        <item name="colorPrimary">@color/purple_500</item>
        <item name="colorPrimaryVariant">@color/purple_700</item>
        <item name="colorOnPrimary">@color/white</item>
        <!-- Secondary brand color. -->
        <item name="colorSecondary">@color/teal_200</item>
        <item name="colorSecondaryVariant">@color/teal_700</item>
        <item name="colorOnSecondary">@color/black</item>
        <!-- Status bar color. -->
        <item name="android:statusBarColor"
tools:targetApi="l">?attr/colorPrimaryVariant</item>
        <!-- Customize your theme here. -->
    </style>

<style name="AppTheme" parent="Theme.AppCompat.Light.DarkActionBar">
    <!-- Customize your theme here. -->
    <item name="colorPrimary">@color/colorPrimary</item>
    <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
    <item name="colorAccent">@color/colorAccent</item>
</style>

</resources>
```

Androidmanifest.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.alarmclock">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.alarmclock">
        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category
android:name="android.intent.category.LAUNCHER" />
```

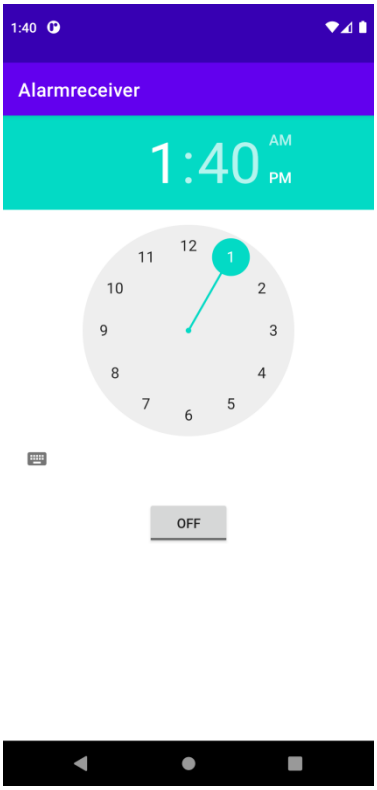
```
            </intent-filter>
        </activity>
    </application>
    <uses-permission android:name="android.permission.VIBRATE" />
</manifest>
```

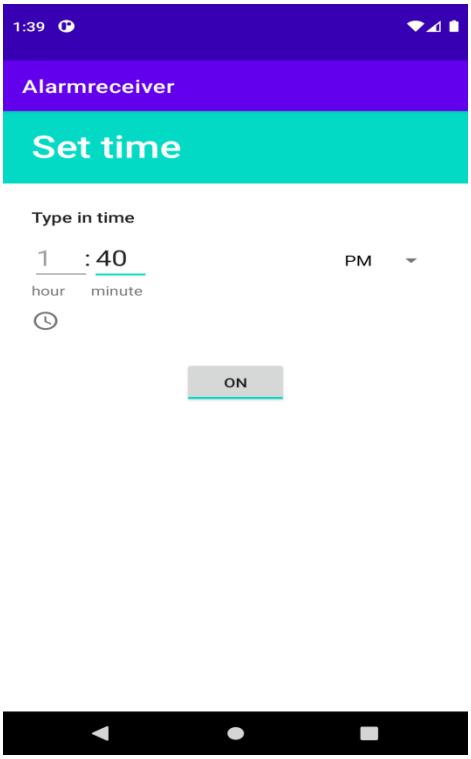
Colors.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="purple_200">#FFBB86FC</color>
    <color name="purple_500">#FF6200EE</color>
    <color name="purple_700">#FF3700B3</color>
    <color name="teal_200">#FF03DAC5</color>
    <color name="teal_700">#FF018786</color>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>


    <color name="colorPrimary">#0F9D58</color>
    <color name="colorPrimaryDark">#0F4C2E</color>
    <color name="colorAccent">#9D0F9B</color>
</resources>
```

OUTPUT:





RESULT:

Thus Android Application that creates Alarm Clock is developed and executed successfully.

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Game - Tic Tac Toe

Expt. No: 10
Date:23/10/2021

Aim:

To develop a simple gaming application with multimedia support

Software and Hardware Requirements:

Software:

1. Android Studio
2. Java

Hardware:

1. Preferably 8GB+ RAM Laptop/Desktop

Algorithm:

1. Open Android Studio and create a new project.
2. Select Empty Activity.
3. Create a function onClick() to add user input as ‘x’ or ‘o’ based on the player.
4. Create a function checkWin() to check if any of the three columns/ rows/ diagonals are having the same value.
5. Create separate functions for player 1 and 2 wins .
6. Within player wins ,add audio using Mediaplayer.
7. Use functions like .start(), . create() and .release() to play and pause the audio.
8. Finally, run the code

Code:

MainActivity.java:

```
package com.example.tictactoe;

import androidx.appcompat.app.AppCompatActivity;

import android.media.MediaPlayer;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.ImageSwitcher;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
```

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```
public class MainActivity extends AppCompatActivity implements
View.OnClickListener {

   private Button[][] buttons = new Button[3][3];

   private boolean player1Turn = true;

   private int roundCount;

   private int player1Points;
   private int player2Points;

   private TextView textViewPlayer1;
   private TextView textViewPlayer2;
   MediaPlayer winner;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       textViewPlayer1 = findViewById(R.id.text_view_p1);
       textViewPlayer2 = findViewById(R.id.text_view_p2);

       for (int i = 0; i < 3; i++) {
           for (int j = 0; j < 3; j++) {
               String buttonID = "button_" + i + j;
               int resID = getResources().getIdentifier(buttonID,
"id", getPackageName());
               buttons[i][j] = findViewById(resID);
               buttons[i][j].setOnClickListener(this);
           }
       }

       Button buttonReset = findViewById(R.id.button_reset);
       buttonReset.setOnClickListener(new View.OnClickListener() {
           @Override
           public void onClick(View v) {
               resetGame();
           }
       });
   }

   @Override
   public void onClick(View v) {
       if (!((Button) v).getText().toString().equals("")) {
           return;
       }
```

```
    if (player1Turn) {
       // textViewPlayer1.setImageResource(R.drawable.yellow);
        ((Button) v).setText("X");
    } else {
        ((Button) v).setText("O");
        //textViewPlayer2.setImageResource(R.drawable.yellow);
    }

    roundCount++;

    if (checkForWin()) {
        if (player1Turn) {
            player1Wins();
        } else {
            player2Wins();
        }
    } else if (roundCount == 9) {
        draw();
    } else {
        player1Turn = !player1Turn;
    }

}

private boolean checkForWin() {
    String[][] field = new String[3][3];

    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            field[i][j] = buttons[i][j].getText().toString();
        }
    }

    for (int i = 0; i < 3; i++) {
        if (field[i][0].equals(field[i][1])
                && field[i][0].equals(field[i][2])
                && !field[i][0].equals("")) {
            return true;
        }
    }

    for (int i = 0; i < 3; i++) {
        if (field[0][i].equals(field[1][i])
                && field[0][i].equals(field[2][i])
                && !field[0][i].equals("")) {
            return true;
        }
```

```
        }

        if (field[0][0].equals(field[1][1])
                && field[0][0].equals(field[2][2])
                && !field[0][0].equals("")) {
            return true;
        }

        if (field[0][2].equals(field[1][1])
                && field[0][2].equals(field[2][0])
                && !field[0][2].equals("")) {
            return true;
        }

        return false;
    }

    private void player1Wins() {
        player1Points++;
        winner = MediaPlayer.create(MainActivity.this,R.raw.win);
        winner.start();
        Toast.makeText(this, "Player 1 wins!",
Toast.LENGTH_SHORT).show();
        updatePointsText();
        //winner.release();
        resetBoard();

    }

    private void player2Wins() {
        player2Points++;
        winner = MediaPlayer.create(MainActivity.this,R.raw.win);
        winner.start();
        Toast.makeText(this, "Player 2 wins!",
Toast.LENGTH_SHORT).show();
        updatePointsText();
        resetBoard();

    }

    private void draw() {
        Toast.makeText(this, "Draw!", Toast.LENGTH_SHORT).show();
        resetBoard();
    }

    private void updatePointsText() {
        textViewPlayer1.setText("Player 1: " + player1Points);
        textViewPlayer2.setText("Player 2: " + player2Points);
```

```
    }

    private void resetBoard() {
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                buttons[i][j].setText("");
            }
        }
        roundCount = 0;
        player1Turn = true;
    }

    private void resetGame() {
        player1Points = 0;
        player2Points = 0;
        updatePointsText();
        resetBoard();
    }

    @Override
    protected void onSaveInstanceState(Bundle outState) {
        super.onSaveInstanceState(outState);

        outState.putInt("roundCount", roundCount);
        outState.putInt("player1Points", player1Points);
        outState.putInt("player2Points", player2Points);
        outState.putBoolean("player1Turn", player1Turn);
    }

    @Override
    protected void onRestoreInstanceState(Bundle savedInstanceState)
{
        super.onRestoreInstanceState(savedInstanceState);

        roundCount = savedInstanceState.getInt("roundCount");
        player1Points = savedInstanceState.getInt("player1Points");
        player2Points = savedInstanceState.getInt("player2Points");
        player1Turn = savedInstanceState.getBoolean("player1Turn");
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
```

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```
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
tools:context="com.example.tictactoe.MainActivity">

<RelativeLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content">
    <TextView
        android:id="@+id/text_view_p1"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:freezesText="true"
        android:text="Player 1: 0"
        android:textSize="30sp" />
    <TextView
        android:id="@+id/text_view_p2"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:freezesText="true"
        android:text="Player 1: 0"
        android:layout_below="@+id/text_view_p1"
        android:textSize="30sp" />


    <Button
        android:id="@+id/button_reset"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentEnd="true"
        android:layout_centerVertical="true"
        android:layout_marginEnd="33dp"
        android:text="reset"
        android:layout_alignParentRight="true"
        android:layout_marginRight="33dp" />

</RelativeLayout>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1">

    <Button
        android:id="@+id/button_00"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1"
```

```
        android:freezesText="true"
        android:textSize="60sp" />

    <Button
        android:id="@+id/button_01"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:freezesText="true"
        android:textSize="60sp" />

    <Button
        android:id="@+id/button_02"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:freezesText="true"
        android:textSize="60sp" />

</LinearLayout>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1">

    <Button
        android:id="@+id/button_10"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:freezesText="true"
        android:textSize="60sp" />

    <Button
        android:id="@+id/button_11"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:freezesText="true"
        android:textSize="60sp" />

    <Button
        android:id="@+id/button_12"
        android:layout_width="0dp"
        android:layout_height="match_parent"
        android:layout_weight="1"
        android:freezesText="true"
```

```
            android:textSize="60sp" />

    </LinearLayout>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1">

        <Button
            android:id="@+id/button_20"
            android:layout_width="0dp"
            android:layout_height="match_parent"
            android:layout_weight="1"
            android:freezesText="true"
            android:textSize="60sp" />

        <Button
            android:id="@+id/button_21"
            android:layout_width="0dp"
            android:layout_height="match_parent"
            android:layout_weight="1"
            android:freezesText="true"
            android:textSize="60sp" />

        <Button
            android:id="@+id/button_22"
            android:layout_width="0dp"
            android:layout_height="match_parent"
            android:layout_weight="1"
            android:freezesText="true"
            android:textSize="60sp" />

    </LinearLayout>

</LinearLayout>
```

SAMPLE I/O :

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RESULT:

Thus a simple gaming application with multimedia support has been build and executed successfully .